F12 4M CAR

Lecture 12: Section 1.9
Inverse Functions

Inverse relation

One-to-one functions

Horizontal line test

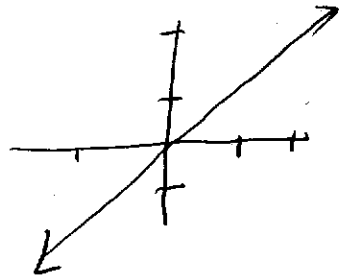
Inverse functions

Properties of inverse functions

Finding inverse functions graphically

Finding inverse functions algebraically

Review: $f(x) = x$ is the line $y = x$



$(f \circ g)(x) = f(g(x))$

Def. If a function f is a set of ordered pairs (x, y), then the **inverse relation** of f is the set of ordered pairs (y, x).

ex. Find the inverse relation of the following functions. Is the inverse a function? → for each x, only one y.

1) $f \ : \ \{(-2, 2), (3, 1), (4, 0), (-1, 2)\}$

$\{(2,-2), (1,3), (0,4), (2,-1)\}$

not a funct.

2) $g \ : \ \{(-2, 2), (3, 1), (0, 0), (4, -1)\}$

$\{(2,-2), (1,3), (0,0), (-1,4)\}$

Funct. - yes

NOTE: An inverse of a function **may not** be a function.

One-to-One Functions

Def. A function f is a **one-to-one** function if each value of the dependent variable (y) corresponds to exactly one value of the independent variable (x).

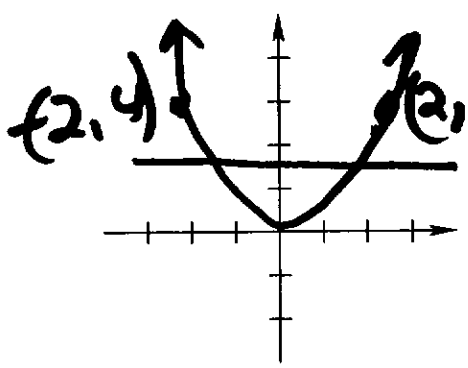
That is, given any two elements x_1 and x_2 in the domain of f, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

NOTE: If a function f is a set of ordered pairs, f is one-to-one if no two ordered pairs have the same second element. Each y has only one x.

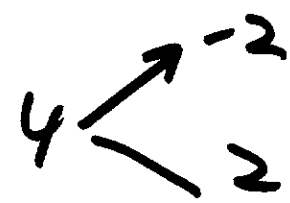
An **inverse** of a function f is also a function if and only if f is a **one-to-one** function.

ex. Let $f(x) = x^2$. Is f a one-to-one function?

Not 1-1.



fails horiz. line test.

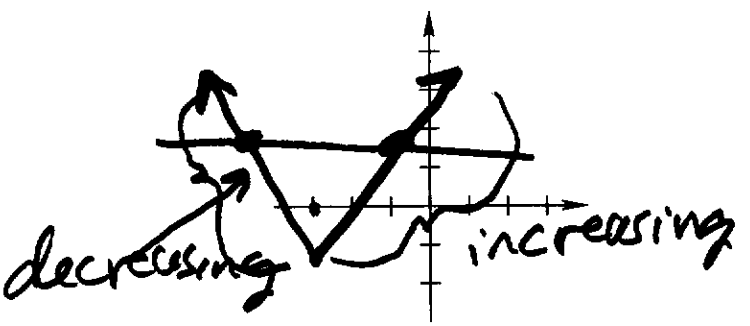


Horizontal Line Test

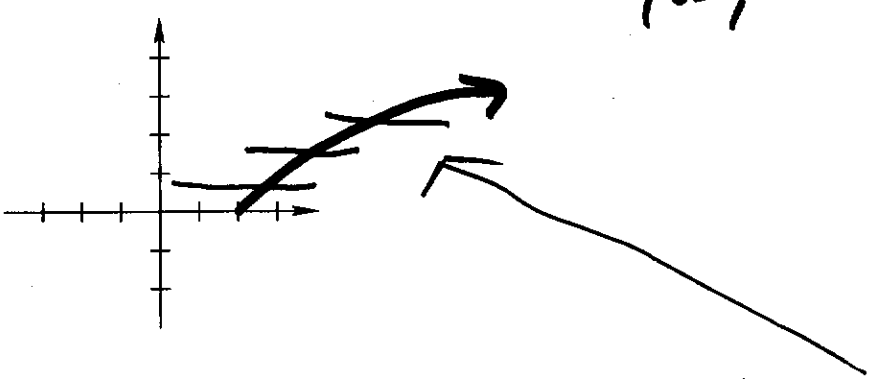
A function f is one-to-one if and only if no horizontal line intersects the graph of f more than once.

<u>**ex.**</u> Determine whether each of the following functions is one-to-one:

1) $f(x) = |x+3| - 1$ Not 1-1.



2) $g(x) = \sqrt{x-2}$ Yes, 1-1.



NOTE: A function that is strictly increasing or decreasing on its domain is one-to-one.

Inverse Functions

Def. let f be a one-to-one function. Then g is the **inverse** function of f if

$(f \circ g)(x) = f(g(x)) = x$ for every x in the domain of g and

$(g \circ f)(x) = g(f(x)) = x$ for every x in the domain of f.

ex. Show that $f(x) = \sqrt[3]{x-1}$ and $g(x) = x^3 + 1$ are inverse functions. Note: Both are 1-1.

$$f(g(x)) = f(x^3+1) = \sqrt[3]{(x^3+1)-1} = \sqrt[3]{x^3} = \boxed{x}$$

$$g(f(x)) = g(\sqrt[3]{x-1}) = (\sqrt[3]{x-1})^3 + 1 = x-1+1$$
$$= \boxed{x}$$

Yes, inverses!

NOTE: If g is the inverse of f, we write $g = f^{-1}$.

We have, $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$

Note: $f^{-1} \neq \frac{1}{f}$

$\frac{1}{f} = (f)^{-1}$

Properties of Inverse Functions

<u>**ex.**</u> If $f(x) = \sqrt[3]{x-1}$, then $f^{-1}(x) = x^3 + 1$.

Find: 1) $f(9) = \sqrt[3]{9-1} = \sqrt[3]{8} = 2$ $(9, 2)$

2) $f^{-1}(2) = 2^3 + 1 = 9$ $(2, 9)$

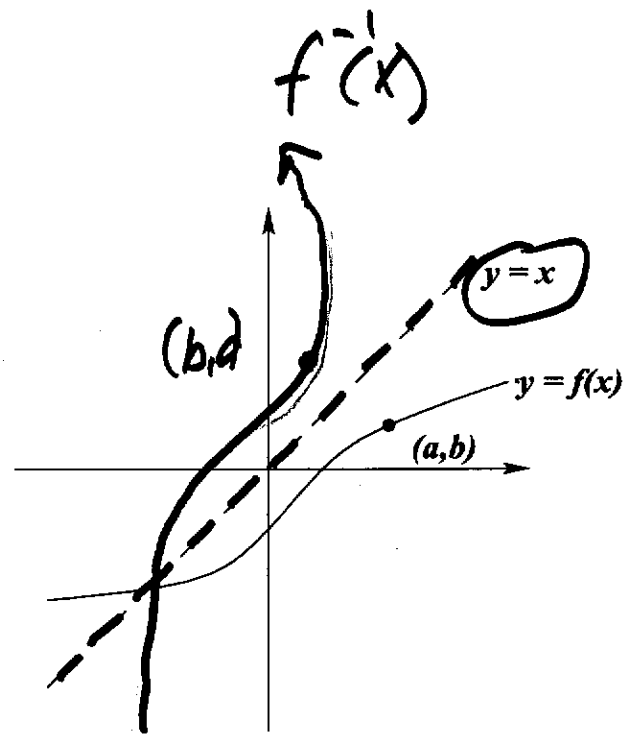
$f: (x, y)$

$f^{-1}: (y, x)$ We have: A point (a, b) is on the graph of one-to-one function f if and only if the point (b, a) is on the graph of f^{-1}.

1) Domain of f = range of f^{-1}

Range of f = domain of f^{-1}

2) The graph of f^{-1} is the reflection of the graph of f through the line $y = x$ (the graph of f and the graph of f^{-1} are symmetric with respect to the line $y = x$).



Finding Inverse Functions Algebraically

1) Use the Horizontal Line Test to make sure the function f is one-to-one.

2) In the equation for $f(x)$, replace $f(x)$ by y.

3) Interchange the variables x and y.

4) Solve for y and replace y by f^{-1} to obtain the explicit function $y = f^{-1}(x)$.

5) Check that the domain of f^{-1} is the range of f and vice versa.

$y = -\frac{1}{3}x + 4$

ex. Find the inverse of $f(x) = 4 - \frac{x}{3}$ if possible.

① Yes, 1-1. straight line with slope $\neq 0$.

② $y = 4 - \frac{x}{3}$

③ $x = 4 - \frac{y}{3}$

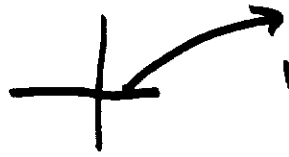
④ $3x = 12 - y$

$y = -3x + 12$

$\boxed{f^{-1}(x) = -3x + 12}$

⑤ $D_f = (-\infty, \infty) = R_{f^{-1}}$

$R_f = (-\infty, \infty) = D_{f^{-1}}$

① 1-1?  yes, 1-1.

② $y = \sqrt{x-1}$

$x - 1 \geq 0$
$x \geq 1$

ex. Find the inverse of $f(x) = \sqrt{x-1}$.

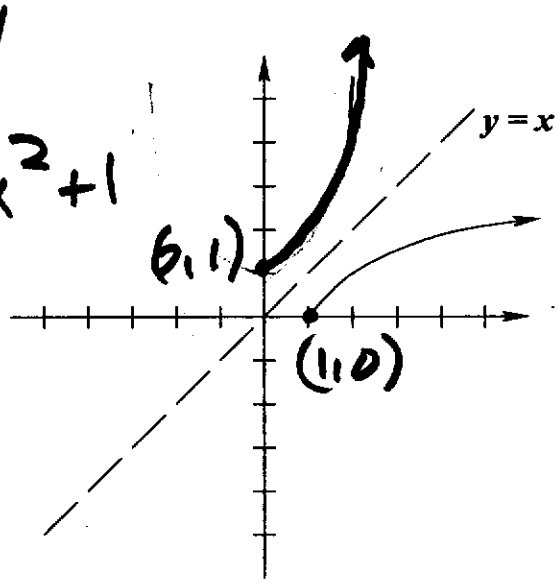
③ $x = \sqrt{y-1}$

④ $x^2 = \left(\sqrt{y-1}\right)^2$

$x^2 = y - 1$

$y = x^2 + 1$

$f^{-1}(x) = x^2 + 1$



⑤ $D_f = [1, \infty) = R_{f^{-1}}$

$R_f = [0, \infty) = D_{f^{-1}}$

thus

$\boxed{f^{-1}(x) = x^2 + 1,\ x \geq 0}$

① Given.

③ $y = \dfrac{2x+1}{x-3}$

ex. Given a one-to-one function $f(x) = \dfrac{2x+1}{x-3}$.

1) Find $f^{-1}(x)$.

③ $x = \dfrac{2y+1}{y-3}$

Solve for y.

④ $x(y-3) = 2y+1$

$xy - 3x = 2y+1$

$xy - 2y = 3x+1$

$y(x-2) = 3x+1$

$y = \dfrac{3x+1}{x-2}$

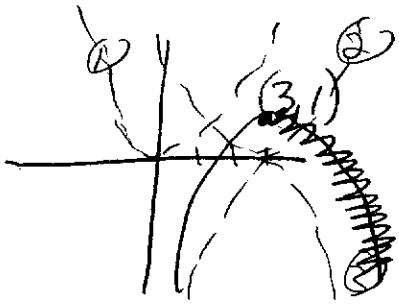
$$\boxed{f^{-1}(x) = \frac{3x+1}{x-2}}$$

2) Find the range of $f(x)$.

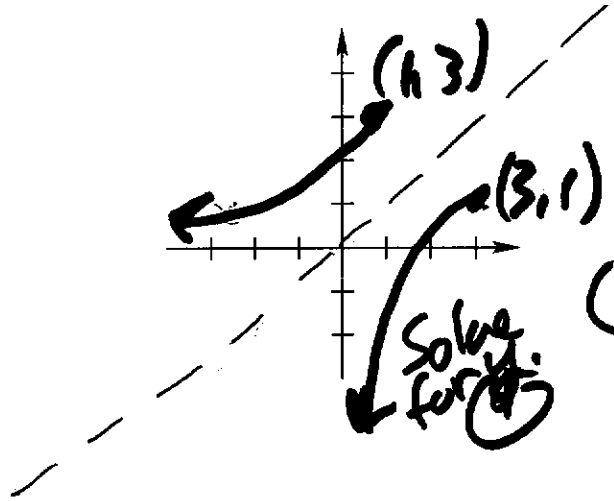
Note: $R_f = D_{f^{-1}}$ so find domain of f^{-1}.

$D_{f^{-1}}$: $x \neq 2$ thus $(-\infty, 2) \cup (2, \infty)$ is the range of f.

① 1-1 ✓



ex. Find the inverse of $f(x) = -(x-3)^2+1$, $x \le 3$.



② $y = -(x-3)^2+1, \quad x \le 3$

③ $x = -(y-3)^2+1, \quad y \le 3$

$x-1 = -(y-3)^2, \quad y \le 3$

$-x+1 = (y-3)^2$

$1-x = (y-3)^2, \quad y \le 3$

Which is it + or − ?

$\pm\sqrt{1-x} = y-3, \quad y \le 3$

Note: $y \le 3$

sub. $y=2$

$2-3$ is neg.

thus −.

$-\sqrt{1-x} = y-3, \quad y \le 3$

$y = 3-\sqrt{1-x}$

$$\boxed{f^{-1}(x) = 3-\sqrt{1-x}}$$

ex Find the inverse of $g(x) = -(x-3)^2+1,\ x \geq 3$

① 1-1 ✓

② $y = -(x-3)^2+1,\ x \geq 3$

③ $x = -(y-3)^2+1,\ y \geq 3$

④ Solve for y.

$x-1 = -(y-3)^2,\quad y \geq 3$

$-x+1 = (y-3)^2,\quad y \geq 3$

$1-x = (y-3)^2,\quad y \geq 3$

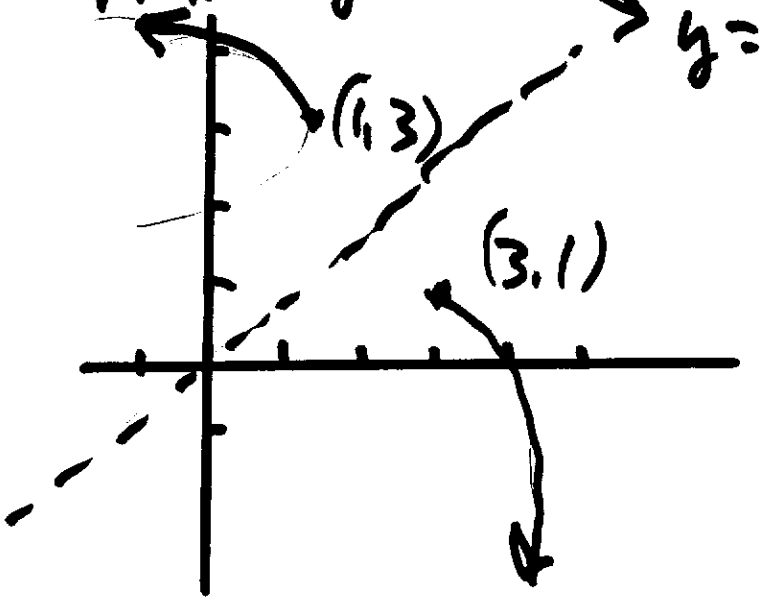
$\pm\sqrt{1-x} = y-3,\quad y \geq 3$

which is it, + or − ?
Use the fact that $y \geq 3$ test 4

$\sqrt{1-x} = y-3$

$y = 3+\sqrt{1-x}$ $4-3 = 1$ is pos.

$\boxed{f^{-1}(x) = 3+\sqrt{1-x}}$



Practice.

Let $f(x) = \sqrt[5]{3x-1}$ and $g(x)$ be a 1-1 function such that $g^{-1}(5) = 2$ and the point $(4, -1)$ lies on the graph of $g(x)$. Evaluate each of the following:

1) $f^{-1}(-1) + g(4)$

2) $f(g(2))$

3) $g(f(11))$

4) $g^{-1}(f(0))$

Answer. 1) -1 2) $\sqrt[5]{14}$ 3) 5 4) 4